

VISHVANATH MADASAMY

Software Engineer | Data Engineering & Analytics | AI & Agentic Workflows

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PROFESSIONAL SUMMARY

Software Engineer specializing in data engineering, analytics platforms, and AI agentic workflows — with hands-on experience building backend APIs, ReactJS UIs, database systems, and LLM-powered intelligent pipelines. I am proficient in Python, ReactJS, MySQL, PostgreSQL, NoSQL, OpenSearch, Docker, Kubernetes, d3.js charting, and microservices architecture. Experienced in designing scalable self-service analytics tools and deploying production-grade AI solutions that transform complex data into actionable business intelligence. AWS Certified AI Practitioner (AIF-01) & Cloud Practitioner (CLF-02). MS Data Science, GPA 3.778/4.0. OPT-eligible May 2026 — willing to relocate to Austin, TX or Fremont, CA.

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript, ReactJS, Java, SQL, C, HTML, CSS — performant coding patterns & scalable software architecture

Backend & APIs: FastAPI, Flask, REST APIs, Microservices architecture, Authentication/Authorization services, API integration, CI/CD pipelines

Databases: MySQL, PostgreSQL, NoSQL (MongoDB), OpenSearch, SQL Server (SSMS), Apache Spark, Hadoop, Hive — schema design, ETL, query optimization

AI & Agentic Workflows: LLMs (GPT-4, Amazon Bedrock, Claude), LangChain, Rasa NLU, PyTorch, Scikit-learn, XGBoost, NLP pipelines, AI-driven analytics automation

Visualization & Orchestration: d3.js, Power BI, Tableau, QuickSight — drill-down charting, interactive dashboards, Airflow data orchestration, self-service BI

DevOps & Cloud: Docker, Kubernetes, Git, AWS (EC2, S3, SageMaker, Lambda, Glue), Azure (Data Factory, Databricks), GCP, Ansible, Chef, Redfish/IPMI

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-01)

Amazon Web Services
Active: Mar 2026 – Mar 2029

AWS Certified Cloud Practitioner (CLF-02)

Amazon Web Services
Active: Mar 2026 – Mar 2029

Microsoft Azure Fundamentals (AZ-900)

In Progress / Pursuing

Microsoft 365 Certified: Administrator Expert (MS-102)

In Progress / Pursuing

WORK EXPERIENCE

University of Massachusetts Dartmouth

North Dartmouth, MA

Information Systems Development & Integration Assistant / IT System Analyst

Sep 2025 – Present

- Designed and developed backend APIs and data pipeline components using Python and microservices architecture — integrating enterprise systems (INFOR POS, UMass Pass, Lenel OnGuard) enabling real-time data access for operational analytics across a 10,000+ user environment.
- Built AI agentic workflows using LLM orchestration (Amazon Bedrock, GPT-4) and LangChain — developed the Log Analyzer tool (Python + PowerShell) to automate log parsing, ML-based anomaly detection, and d3-style geolocation visualization of security threats across the network.
- Developed ReactJS-based UI components for internal analytics dashboards — implementing intuitive data rendering, drill-down charting, and interactive visualizations enabling self-service business intelligence for non-technical stakeholders.
- Containerized and deployed applications using Docker and Kubernetes — managing CI/CD pipelines, Ansible/Chef configuration management, and automated testing workflows ensuring reliable, scalable production deployments.
- Administered MySQL, PostgreSQL, NoSQL (MongoDB), and OpenSearch database systems — designing schemas, optimizing queries, and building ETL pipelines enabling real-time analytics and reporting platform functionality.
- Collaborated with architects to define AI/analytics roadmap and drove Cursor AI adoption across development workflows owing full POC-to-production lifecycle and guiding team members on best practices.

Value IBMS Pvt Ltd.

Chennai, India

Data Analyst Intern

Oct 2023 – Dec 2023

- Architected and deployed a production-grade Python + SQL Server ETL pipeline on cloud infrastructure — automating end-to-end data ingestion, transformation, and loading workflows that reduced data processing time by 45% and unlocked real-time analytics for business teams.
- Built interactive Power BI and Tableau KPI dashboards with drill-down analytics, ML-based trend forecasting, and Excel/CSV automated reporting — delivering actionable business intelligence that directly drove a 30% increase in product sales.
- Designed microservice-based data ingestion workflows integrating REST APIs with SQL Server backend — enabling real-time reporting pipelines and self-service analytics for cross-functional business stakeholders.
- Applied Python (Pandas, NumPy) and Machine Learning models for data cleaning, preprocessing, and predictive analysis — improving data quality and enabling proactive decision-making across sales and operations teams.

Value IBMS Pvt Ltd.

Chennai, India

IT System Analyst Intern

Jul 2023 – Aug 2023

- Deployed automated ML-based monitoring and analytics solution for Pune Metro CPM infrastructure — achieving 37% improvement in building management system efficiency through intelligent data-driven alerting and visualization.
- Designed and deployed an automated ML-based monitoring and analytics solution for the Chiller Plant Manager (CPM) project at Pune Metro Station — integrating IoT sensor data streams with Python automation scripts to achieve a 37% improvement in building management system efficiency.
- Built Python automation scripts for real-time IoT data ingestion, SQL-based query optimization, and automated report generation — eliminating manual monitoring processes and reducing response time to infrastructure anomalies.
- Performed end-to-end data pipeline work including IoT sensor data collection, cleaning, transformation, and visualization — ensuring high-quality, reliable data for engineering and operations decision-making.

KEY PROJECTS

AI-Powered Medical Image Analysis Platform | Python, Next.js, ReactJS, Google Cloud | [GitHub](#)

Oct 2025

- Built a full-stack production-grade AI analytics platform with ReactJS frontend, Python backend APIs, LLM-powered inference pipeline, and GCP cloud deployment — demonstrating end-to-end data engineering, agentic AI, and scalable cloud architecture capability.

Global Microplastics Network Visualization | Python, d3.js, Visual Analytics | [GitHub](#)

May 2025

- Developed an interactive d3.js network visualization system enabling clickable node-based exploration of worldwide microplastic pollution data — implementing drill-down analytics, geographic concentration statistics, and real-time data rendering for scientific decision-making.

Car Rental Management System | Python, ReactJS, HTML/CSS, MySQL, SSMS | Database Design Coursework | [Github](#)

May 2025

- Designed and developed a full-stack system with Python backend, ReactJS/HTML/CSS frontend, and MySQL/SSMS database — implementing CRUD operations, real-time availability querying, dynamic pricing logic, and an admin analytics dashboard for fleet management.

Library Assistant Chatbot — Conversational AI | Python, Rasa NLU, Flask, REST API |

Jul 2024

- Designed and deployed an NLP agentic chatbot using Rasa NLU and Flask REST API — implementing intent classification, entity recognition, dialogue management, and webhook integration for fully automated self-service support workflows.

EDUCATION

Master of Science in Data Science | GPA: 3.778 / 4.0

Aug 2024 – May 2026

University of Massachusetts Dartmouth — North Dartmouth, MA

Coursework: AI, Advanced ML, Database Design, Network Security, High Performance Scientific Computation, Data Mining, Statistical Analysis, Data Visualization

Bachelor of Engineering — Electronics & Communication Engineering | GPA: 8.65 / 10

Jul 2020 – Apr 2024

Anna University — Chennai, India

Best Undergraduate Final Project Award 2024 — ML-based automated image detection system